

Regression Analysis: Transmission Architecture and Fuel Efficiency

Motor Trend Analyst

2026-06-06

Executive Summary

This analysis evaluates the structural impact of transmission type (automatic vs. manual) on vehicle fuel efficiency (MPG) using the mtcars dataset. Exploratory analysis indicates a significant fuel economy advantage for manual transmissions (+7.245 MPG). However, OLS modeling reveals that this variance is due to vehicle weight (wt) and horsepower (hp). After isolating these attributes via nested model selection and Likelihood Ratio Testing, manual transmissions are associated with an increase of 2.084 MPG over automatic transmissions when weight and horsepower are constant. This adjusted effect is not statistically significant at the $\alpha = 0.05$ threshold ($p = 0.141$), indicating that weight and horsepower are the primary drivers of fuel efficiency, not transmission type.

Exploratory Data Analysis

The sample mean for manual vehicles is 24.39 MPG while automatic vehicles average 17.15 MPG. A two-sample Welch t-test confirms a statistically significant unadjusted difference:

```
t_res <- t.test(mpg ~ am, data = mtcars)
t_res

#>
#> Welch Two Sample t-test
#>
#> data: mpg by am
#> t = -3.7671, df = 18.332, p-value = 0.001374
#> alternative hypothesis: true difference in means is not equal to 0
#> 95 percent confidence interval:
#> -11.280194 -3.209684
#> sample estimates:
#> mean in group Automatic    mean in group Manual
#>          17.14737          24.39231
```

The difference is 7.24 MPG ($t(18.33) = -3.77$, $p < 0.005$). Fig 1 shows automatic transmissions are clustered heavily in high-wt, high-hp vehicles.

Model Selection Strategy & Variable Pruning

Model 1: Regress MPG strictly on transmission type (am).

Model 2: Introduce wt and hp.

Model 3: Stepwise AIC selection to scan variables.

```
fit_base <- lm(mpg ~ am, data = mtcars)
fit_ctrl <- lm(mpg ~ am + wt + hp, data = mtcars)
fit_full <- stepAIC(lm(mpg ~ ., data = mtcars), direction = "both")

#> Start: AIC=76.4
#> mpg ~ cyl + disp + hp + drat + wt + qsec + vs + am + gear + carb
```

```

#>
#>      Df Sum of Sq  RSS   AIC
#> - carb  5   13.5989 134.00 69.828
#> - gear  2    3.9729 124.38 73.442
#> - am    1    1.1420 121.55 74.705
#> - qsec  1    1.2413 121.64 74.732
#> - drat  1    1.8208 122.22 74.884
#> - cyl   2   10.9314 131.33 75.184
#> - vs    1    3.6299 124.03 75.354
#> <none>          120.40 76.403
#> - disp  1    9.9672 130.37 76.948
#> - wt    1   25.5541 145.96 80.562
#> - hp    1   25.6715 146.07 80.588
#>
#> Step:  AIC=69.83
#> mpg ~ cyl + disp + hp + drat + wt + qsec + vs + am + gear
#>
#>      Df Sum of Sq  RSS   AIC
#> - gear  2    5.0215 139.02 67.005
#> - disp  1    0.9934 135.00 68.064
#> - drat  1    1.1854 135.19 68.110
#> - vs    1    3.6763 137.68 68.694
#> - cyl   2   12.5642 146.57 68.696
#> - qsec  1    5.2634 139.26 69.061
#> <none>          134.00 69.828
#> - am    1   11.9255 145.93 70.556
#> - wt    1   19.7963 153.80 72.237
#> - hp    1   22.7935 156.79 72.855
#> + carb  5   13.5989 120.40 76.403
#>
#> Step:  AIC=67
#> mpg ~ cyl + disp + hp + drat + wt + qsec + vs + am
#>
#>      Df Sum of Sq  RSS   AIC
#> - drat  1    0.9672 139.99 65.227
#> - cyl   2   10.4247 149.45 65.319
#> - disp  1    1.5483 140.57 65.359
#> - vs    1    2.1829 141.21 65.503
#> - qsec  1    3.6324 142.66 65.830
#> <none>          139.02 67.005
#> - am    1   16.5665 155.59 68.608
#> - hp    1   18.1768 157.20 68.937
#> + gear  2    5.0215 134.00 69.828
#> - wt    1   31.1896 170.21 71.482
#> + carb  5   14.6475 124.38 73.442
#>
#> Step:  AIC=65.23
#> mpg ~ cyl + disp + hp + wt + qsec + vs + am
#>
#>      Df Sum of Sq  RSS   AIC
#> - disp  1    1.2474 141.24 63.511
#> - vs    1    2.3403 142.33 63.757
#> - cyl   2   12.3267 152.32 63.927
#> - qsec  1    3.1000 143.09 63.928

```

```

#> <none>                139.99 65.227
#> + drat  1      0.9672 139.02 67.005
#> - hp    1     17.7382 157.73 67.044
#> - am    1     19.4660 159.46 67.393
#> + gear  2      4.8033 135.19 68.110
#> - wt    1     30.7151 170.71 69.574
#> + carb  5     13.0509 126.94 72.095
#>
#> Step:  AIC=63.51
#> mpg ~ cyl + hp + wt + qsec + vs + am
#>
#>      Df Sum of Sq  RSS   AIC
#> - qsec  1      2.442 143.68 62.059
#> - vs    1      2.744 143.98 62.126
#> - cyl   2     18.580 159.82 63.466
#> <none>                141.24 63.511
#> + disp  1      1.247 139.99 65.227
#> + drat  1      0.666 140.57 65.359
#> - hp    1     18.184 159.42 65.386
#> - am    1     18.885 160.12 65.527
#> + gear  2      4.684 136.55 66.431
#> - wt    1     39.645 180.88 69.428
#> + carb  5      2.331 138.91 72.978
#>
#> Step:  AIC=62.06
#> mpg ~ cyl + hp + wt + vs + am
#>
#>      Df Sum of Sq  RSS   AIC
#> - vs    1      7.346 151.03 61.655
#> <none>                143.68 62.059
#> - cyl   2     25.284 168.96 63.246
#> + qsec  1      2.442 141.24 63.511
#> - am    1     16.443 160.12 63.527
#> + disp  1      0.589 143.09 63.928
#> + drat  1      0.330 143.35 63.986
#> + gear  2      3.437 140.24 65.284
#> - hp    1     36.344 180.02 67.275
#> - wt    1     41.088 184.77 68.108
#> + carb  5      3.480 140.20 71.275
#>
#> Step:  AIC=61.65
#> mpg ~ cyl + hp + wt + am
#>
#>      Df Sum of Sq  RSS   AIC
#> <none>                151.03 61.655
#> - am    1      9.752 160.78 61.657
#> + vs    1      7.346 143.68 62.059
#> + qsec  1      7.044 143.98 62.126
#> - cyl   2     29.265 180.29 63.323
#> + disp  1      0.617 150.41 63.524
#> + drat  1      0.220 150.81 63.608
#> + gear  2      1.361 149.66 65.365
#> - hp    1     31.943 182.97 65.794
#> - wt    1     46.173 197.20 68.191

```

```
#> + carb 5      5.633 145.39 70.438
```

Evaluate whether the complexity of Model 3 is justified over Model 2 using ANOVA analysis:

```
anova(fit_ctrl, fit_full)
```

```
#> Analysis of Variance Table
#>
#> Model 1: mpg ~ am + wt + hp
#> Model 2: mpg ~ cyl + hp + wt + am
#>   Res.Df    RSS Df Sum of Sq    F Pr(>F)
#> 1      28 180.29
#> 2      26 151.03  2    29.265 2.5191   0.1 .
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The test yields an p-value of approximately 0.10. We fail to reject the null hypothesis (the additional parameters in the full model are zero).

Model 2 is optimal for inference.

Statistical Inference and Parameter Interpretation

Parameter coefficients:

```
summary(fit_ctrl)$coefficients
```

```
#>               Estimate Std. Error  t value    Pr(>|t|)
#> (Intercept) 34.00287512  2.642659337 12.866916 2.824030e-13
#> amManual     2.08371013  1.376420152  1.513862 1.412682e-01
#> wt          -2.87857541  0.904970538 -3.180850 3.574031e-03
#> hp          -0.03747873  0.009605422 -3.901830 5.464023e-04
```

Interpretation of Coefficients:

$\hat{\beta}_0$ (intercept = 34.0028): The theoretical expected fuel economy for an automatic vehicle weighing 0 lbs with 0 horsepower.

$\hat{\beta}_{Manual}$ (am = 2.0837): Holding vehicle weight and horsepower perfectly constant, a manual transmission is expected to yield an increase of 2.084 MPG over an automatic transmission.

$\hat{\beta}_{wt}$ (wt = -2.8786): Holding transmission type and horsepower constant, every 1,000 lb increase in vehicle mass reduces efficiency by 2.879 MPG.

Uncertainty Quantification

95% confidence intervals for our model parameters:

```
confint(fit_ctrl)["amManual", ]
```

```
#>      2.5 %      97.5 %
#> -0.7357587  4.9031790
```

The adjusted effect of a manual transmission lies between -0.74 MPG and 4.90 MPG. Because this interval captures zero and the corresponding partial t-test p-value is 0.141, we conclude that the transmission type does not have a statistically significant effect on fuel economy once structural engine and mass metrics are equalized.

Appendix:

Figure 1: Exploratory Matrix Profiling

```

par(mfrow = c(1, 2))
boxplot(mpg ~ am, data = mtcars, col = c("lightcoral", "lightgreen"),
        xlab = "Transmission Type", ylab = "Miles Per Gallon (MPG)",
        main = "Unadjusted MPG by Transmission")
plot(mpg ~ wt, data = mtcars, col = as.numeric(mtcars$am) + 1, pch = 19,
     xlab = "Weight (1,000 lbs)", ylab = "MPG", main = "Confounding: MPG vs Weight")
legend("topright", legend = c("Automatic", "Manual"), col = c(2, 3), pch = 19)

```

Unadjusted MPG by Transmissi

Confounding: MPG vs Weigh

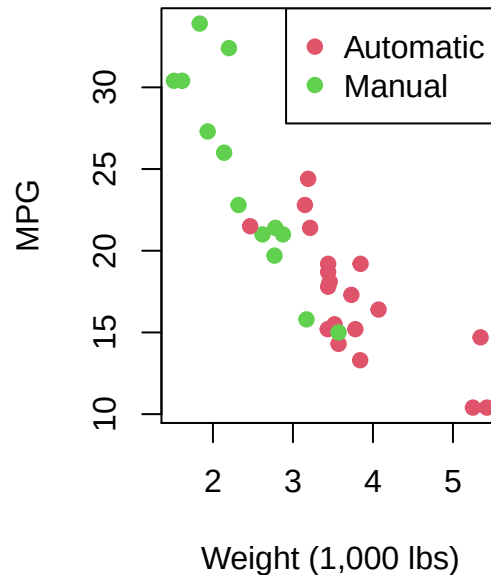
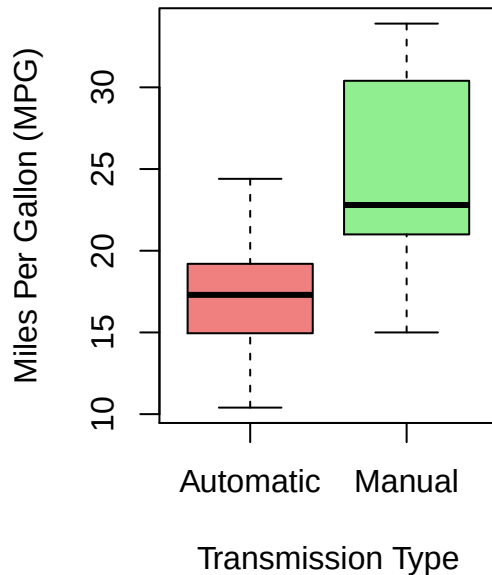
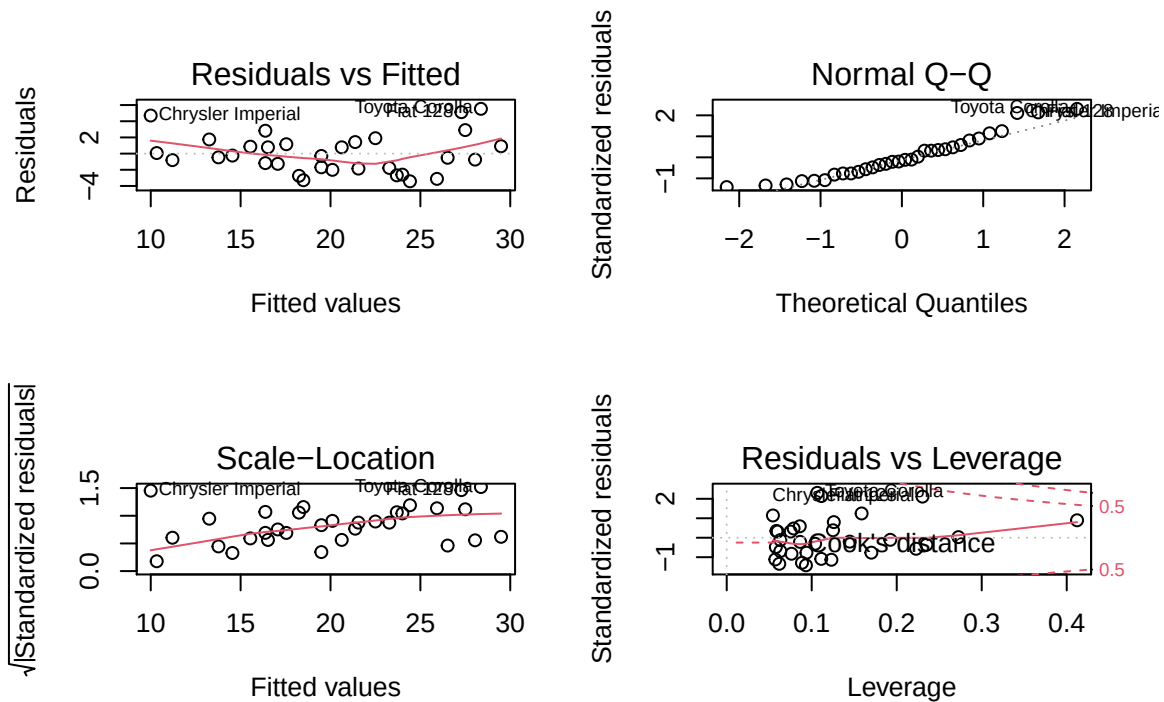


Figure 2: Residual Diagnostics and Regression Surface Auditing

```

par(mfrow = c(2, 2))
plot(fit_ctrl)

```



Diagnostics Assessment:

Residuals vs Fitted: The points bounce around the horizontal zero line, indicating that the assumption of a linear structural relationship is valid.

Normal Q-Q: The standardized residuals adhere closely to the theoretical line, verifying that the error terms are normally distributed.

Scale-Location: The constant vertical spread of the residuals confirms homoskedastic error variance.

Residuals vs Leverage: No data points breach Cook's Distance contours. The single high-leverage asset (the Chrysler Imperial) does not display an extreme standardized residual, confirming our parameters are safe from outlier distortion.